

Lab03_Mazo

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Questão 1

```
library(readr)
library(dplyr)
```

Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

filter, lag

Os seguintes objetos são mascarados por 'package:base':

intersect, setdiff, setequal, union

```
library(stringr)
library(tidyverse)
```

-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --

v forcats	1.0.1	v purrr	1.2.2
v ggplot2	4.0.3	v tibble	3.3.1
v lubridate	1.9.5	v tidyr	1.3.2

-- Conflicts ----- tidyverse_conflicts() --

x dplyr::filter() masks stats::filter()
x dplyr::lag() masks stats::lag()

i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become

```
library(webshot2)

airports <- read_csv(
  "~/Lab03 ME315/airports.csv",
  col_select = c(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE)
)
```

Rows: 322 Columns: 5

-- Column specification -----

Delimiter: ","

chr (3): IATA_CODE, CITY, STATE

dbl (2): LATITUDE, LONGITUDE

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
airlines <- read_csv("~/Lab03 ME315/airlines.csv")
```

Rows: 14 Columns: 2

-- Column specification -----

Delimiter: ","

chr (2): IATA_CODE, AIRLINE

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
estadisticas <- function(x, pos) {

  x <- x |>
  filter(
    !is.na(DESTINATION_AIRPORT),
    !is.na(ARRIVAL_DELAY)
  )

  aeroportos_validos <- str_subset(
    unique(x$DESTINATION_AIRPORT),
    "^[^1]"
  )

  x <- x |>
  filter(DESTINATION_AIRPORT %in% aeroportos_validos)
```

```

x |> group_by(DESTINATION_AIRPORT) |>
  summarise(
    n = n(),
    soma_atraso = sum(ARRIVAL_DELAY),
    .groups = "drop"
  )
}

resultado <- read_csv_chunked(
  "~/Lab03 ME315/flights.csv",
  callback = DataFrameCallback$new(estatisticas),
  chunk_size = 1000000,
  col_types = cols_only(
    DESTINATION_AIRPORT = col_character(),
    ARRIVAL_DELAY = col_double()
  )
)

estatisticas_finais <- bind_rows(resultado) |>
  group_by(DESTINATION_AIRPORT) |>
  summarise(
    n = sum(n),
    soma_atraso = sum(soma_atraso),
    .groups = "drop"
  )

flights <- estatisticas_finais |>
  mutate(media_atraso = soma_atraso / n) |>
  select(
    DESTINATION_AIRPORT,
    media_atraso
  )

View(flights)

```

Questão 2

Resposta:

Chave na tabela flights: DESTINATION_AIRPORT

Chave na tabela airports: IATA_CODE

```
aeroportos_ausentes <- flights |>
  anti_join(
    airports,
    by = c("DESTINATION_AIRPORT" = "IATA_CODE")
  ) |>
  distinct(DESTINATION_AIRPORT)

print(aeroportos_ausentes)
```

```
# A tibble: 0 x 1
# i 1 variable: DESTINATION_AIRPORT <chr>
```

```
flights <- flights |>
  left_join(
    airports |>
      select(IATA_CODE, CITY, STATE),
    by = c("DESTINATION_AIRPORT" = "IATA_CODE")
  )

flights
```

```
# A tibble: 322 x 4
  DESTINATION_AIRPORT media_atraso CITY      STATE
  <chr>                <dbl> <chr>    <chr>
1 ABE                  5.80 Allentown PA
2 ABI                  4.08 Abilene   TX
3 ABQ                  5.61 Albuquerque NM
4 ABR                 -3.39 Aberdeen  SD
5 ABY                  8.68 Albany   GA
6 ACK                  4.91 Nantucket MA
7 ACT                 10.6 Waco      TX
8 ACV                  9.68 Arcata/Eureka CA
9 ACY                 11.8 Atlantic City NJ
10 ADK                 -3.19 Adak     AK
# i 312 more rows
```

Questão 3

```
aeroportos_por_estado <- flights |>
  count(STATE, name = "numero_aeroportos") |>
  arrange(desc(numero_aeroportos))
```

```
aeroportos_por_estado
```

```
# A tibble: 54 x 2
  STATE numero_aeroportos
  <chr>         <int>
1 TX             24
2 CA             22
3 AK             19
4 FL             17
5 MI             15
6 NY             14
7 CO             10
8 MN              8
9 MT              8
10 NC             8
# i 44 more rows
```

Questão 4

```
library(leaflet)

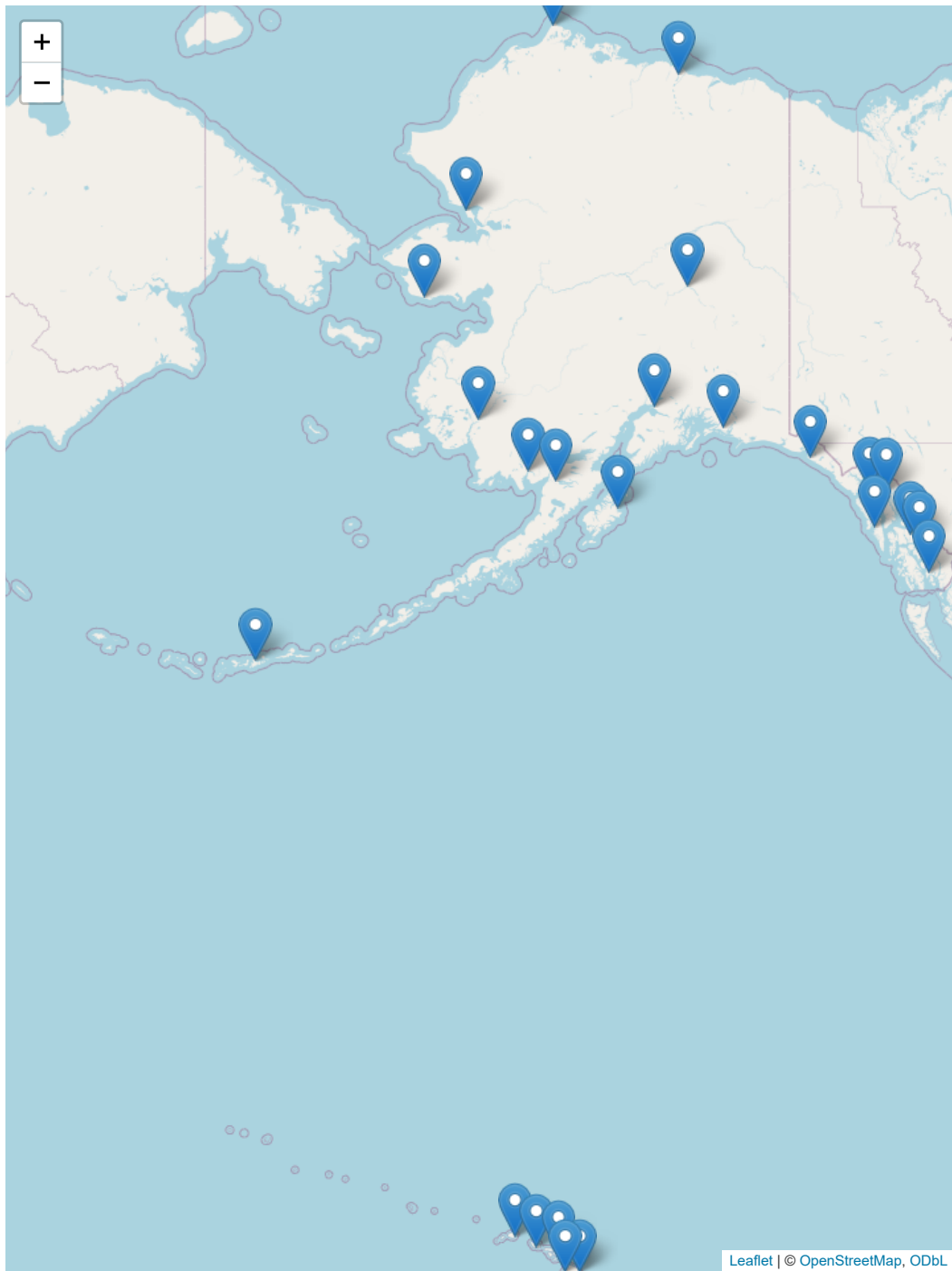
this <- flights |>
  left_join(airports |>
    select(IATA_CODE, LATITUDE, LONGITUDE),
    by = c("DESTINATION_AIRPORT" = "IATA_CODE")
  ) |>
  leaflet() |>
  addTiles() |>
  addMarkers(
    lng = ~LONGITUDE,
    lat = ~LATITUDE,
    popup = ~paste0(
      "<b>Aeroporto:</b> ", DESTINATION_AIRPORT,
      "<br><b>Cidade:</b> ", CITY,
      "<br><b>Estado:</b> ", STATE,
```

```
    "<br><b>Atraso médio:</b> ", round(media_atraso, 2), " minutos"
  )
)
```

Warning in validateCoords(lng, lat, funcName): Data contains 3 rows with either missing or invalid lat/lon values and will be ignored

```
this
```

Warning in file.info(file): cannot resolve owner of file
'\\smb\ra187263\Downloads\file5fecae33c53\webshot5fec5a697c07.pdf': Não foi
feito mapeamento entre os nomes de conta e as identificações de segurança



Horário

```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-08-31 18:42:58 -03"
```